SECTION A

*Write about a quarter of a page each on any **four** of the following topics.*

5. *ETS and ARIMA models are largely interchangeable, so it does not matter which one we use.*

- These model classes are not interchangeable. 1
- Some of the simpler additive models are equivalent including SES and Holt's method with additive errors. 1
- ETS models with multiplicative errors handle heteroskedasticity, whereas ARIMA models do not. 1
- ARIMA models handle a much wider range of dynamics than ETS models. 1
- ETS models cannot be used for stationary series. 1

SECTION B

(g) *ARIMA(2,4,2)(1,1,0)$_7$ applied to the log transformed data.*

- Unsuitable. Far too many differences. 1

(h) *ARIMA(1,0,1)(1,1,0)$_7$ applied to the log transformed data.*

- Might be suitable, but will miss annual seasonality and holidays. 1

SECTION D

1. *An ARIMA model is fitted to the time series, using only data from March 2022. Write down the equation for the model using backshift notation, including specifying the values for all model parameters.* 3

$$(1-\phi_1 B)(1-\Phi_1 B^7)(1-B^7)y_t = (1+\theta_1 B)\varepsilon_t$$

where $y_t = \log(\text{Boardings})$, $\phi_1 = 0.8165$, $\theta_1 = -0.5920$, $\Phi_1 = -0.4574$, $\varepsilon_t \sim N(0, 0.1045)$. 2

2. *This model suggests that the weekly differences of the logged data are stationary. What aspects of the model lead to that conclusion?*

   - The fitted model has $D = 1$ for weekly differences, but $d = 0$ so no ordinary differencing, and it is applied to the logged data. So if we apply these differences to the logged data, we should have stationary data. 3

3. *Let the observed data be given by y_t, and define $x_t = \log(y_t) - \log(y_{t-7})$. The x_t series is shown in Figure 5. What features of these plots suggest that x_t is stationary?*

   - The ACF and PACF both decay to zero quickly. 1
   - The time plot is consistent with a zero mean and constant variance. 1

4. *The residuals shown in Figure 6 show a lot of large outliers. What might be causing these outliers?*

   - These are related to holidays, or the start and end of holiday periods, as these have not been captured by the model. 2

5. *The residuals shown in Figure 6 are clearly not white noise, and are not normally distributed. What features of the plots suggest this?*

   - The ACF shows significant serial correlation, so not white noise. 1
   - The time plot shows large outliers, so not normally distributed. 1

6. *If you produced forecasts from this model, how reliable do you think the point forecasts and prediction intervals would be? Why?*

   - The point forecasts will be biased around holiday times, as the model has not captured the holiday effects. 1
   - The prediction intervals may have inaccurate coverage due to the non-normality of residuals, and the serial correlation present in the residuals. 2

7. *Another model is found with 10 parameters, a slightly lower AICc value, but with no improvement in the residuals. Would you prefer this model to the one you have already fitted? Why or why not?*

   - No. Small differences in AICc don't matter much, and it doesn't solve the problems of the existing model. 2
   - Also a model with a large number of parameters is more difficult to estimate. 1